

LEVEL 1: EASY (Identification & Basic Zero Rules)

Exercise 1

Determine the number of significant figures
in:

0.007040

Answer 1

1. Leading zeros (0.00) are NOT significant.
2. Non-zero digits (7,4) are significant.
3. Captive zero between 7 and 4 is significant.
4. Trailing zero after decimal is significant.

= 4 Significant Figures

Exercise 2

Round 45.8926 to 3 significant figures:

45.8926

Answer 2

1. Identify the first 3 significant digits: 4, 5, 8.
2. Look at the 4th digit (9), which is ≥ 5 .
3. Round up the 3rd digit from 8 to 9.

= 45.9

Exercise 3

Express 68000 in scientific notation with 3 significant figures:

68000

Answer 3

1. Move decimal 4 places left: 6.8×10^4 .
2. Add a trailing zero after decimal to make it 3 sig figs:

= 6.80×10^4

LEVEL 2: INTERMEDIATE (Operations with Precision Rules)

Exercise 4

Perform calculation to the correct number of sig figs:

$$3.24 \times 1.2$$

Answer 4

1. Raw product: $3.24 \times 1.2 = 3.888$. 2. 3.24 has 3 sig figs; 1.2 has 2 sig figs. 3. Limit answer to 2 sig figs (fewest).

$$= 3.9$$

Exercise 5

Perform calculation to correct decimal precision:

$$14.215 + 3.12$$

Answer 5

1. Raw sum: $14.215 + 3.12 = 17.335$. 2. 14.215 has 3 decimal places; 3.12 has 2 decimal places. 3. Round to 2 decimal places (least precise).

$$= 17.34$$

Exercise 6

Calculate volume of a box: $L = 5.0$ cm,
 $W = 2.25$ cm, $H = 1.10$ cm:

$$V = L \times W \times H$$

Answer 6

1. $V = 5.0 \times 2.25 \times 1.10 = 12.375$ cm³. 2. Lowest sig figs among terms is 2 (from 5.0). 3. Round answer to 2 sig figs.

$$= 12 \text{ cm}^3$$

Exercise 7

Calculate density: Mass = 48.2 g, Volume = 6.0 cm³:

$$D = \frac{\text{Mass}}{\text{Volume}}$$

Answer 7

1. $D = 48.2 \div 6.0 = 8.0333... \text{ g/cm}^3$. 2. Mass has 3 sig figs, Volume has 2 sig figs. 3. Round answer to 2 sig figs.

$$= \mathbf{8.0 \text{ g/cm}^3}$$

LEVEL 3: HARD (Combined Operations & Multi-Step Logic)

Exercise 8

Evaluate with proper sig figs:

$$\frac{25.42 - 18.1}{2.5}$$

Answer 8

1. Subtraction step: $25.42 - 18.1 = 7.32 \rightarrow 7.3$ (1 decimal place = 2 sig figs). 2. Division step: $7.3 \div 2.5 = 2.92$. 3. Both terms have 2 sig figs, so round answer to 2 sig figs.

$$= 2.9$$

Exercise 9

Evaluate with proper sig figs:

$$(1.20 \times 10^3) + (4.5 \times 10^2)$$

Answer 9

1. Convert to same power of 10: $1200 + 450 = 1650$. 2. Or as decimals: $1.20 \times 10^3 + 0.45 \times 10^3 = 1.65 \times 10^3$. 3. Limit by least precise place (hundreds place from 1.20×10^3):

$$= 1.65 \times 10^3$$

Exercise 10

A student measures dimensions: $a = 12.1$ cm, $b = 0.52$ cm, $c = 100$ cm. Calculate $a+b+c$:

$$a + b + c$$

Answer 10

1. $12.1 + 0.52 + 100 = 112.62$. 2. Precision limit: 100 (no decimal, ambiguous to hundreds place, or ones place if exact). Assuming measured 100 with no decimal point, least precise digit is in the hundreds place. 3. Rounding gives 100 or if written clearly 113 (if 100 is considered exact count). Assuming measured value:

$$= \mathbf{113} \text{ (or } \mathbf{1.13 \times 10^2})$$

Exercise 11

Convert 0.000045090 m to scientific notation preserving exact sig figs:

$$0.000045090 \text{ m}$$

Answer 11

1. Count sig figs: 4, 5, 0, 9, 0 $\Rightarrow$ 5 sig figs. 2. Move decimal 5 places right: 4.5090. 3. Keep trailing zero to maintain 5 sig figs.

$$= \mathbf{4.5090 \times 10^{-5} \text{ m}}$$